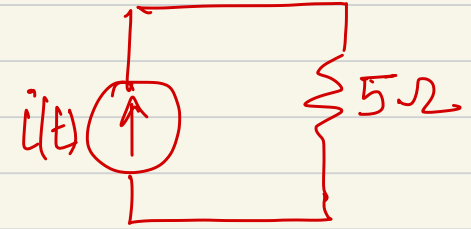


Assessment-I Solutions

Q1. $i(t) = 4 + 2 \cos(2\pi t)$ A



A. RMS value of the current signal.

$$\begin{aligned} I_{RMS} &= \sqrt{\frac{1}{T} \int_0^T i^2(t) dt} \\ &= \sqrt{\frac{1}{T} \int_0^T (16 + 16 \cos(2\pi t) + 4 \cos^2(2\pi t)) dt} \\ &= \sqrt{16 + \frac{1}{T} \int_0^T 16 \cos(2\pi t) dt + \frac{4}{T} \int_0^T \frac{1 + \cos(4\pi t)}{2} dt} \\ &= \sqrt{16 + \frac{16}{T} \times 0 + \frac{2}{T} \times T - \frac{2}{T} \times 0} \\ I_{RMS} &= \sqrt{18} \text{ A} \end{aligned}$$

B. Average power delivered to the resistor τ

$$= \frac{1}{T} \int_0^T i^2(t) \cdot 5 \, dt$$

$$= \frac{5}{T} \int_0^T (4 + 2 \cos(2\pi t))^2 \, dt$$

$$= 5 \times 18 = 90 \text{ W}$$

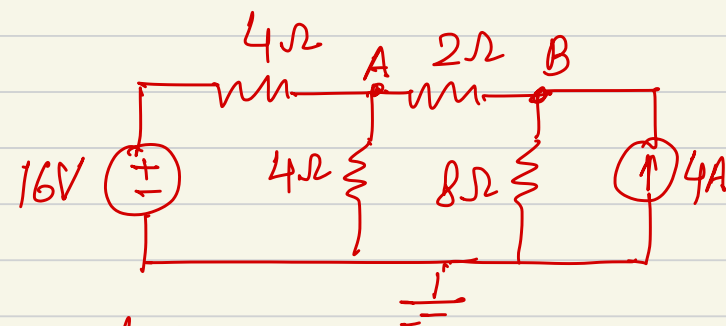
or directly, $P_{av} = I_{RMS}^2 R$

Q2. A. Power dissipated in 2Ω resistor

We can use multiple analysis methods here. As the part B will require

us to know the current direct to comment,

we can go for nodal/mesh analysis to calculate node voltages/current in branches.



Applying KCL at node A :- $\frac{V_A - 16}{4} + \frac{V_A}{4} + \frac{V_A - V_B}{2} = 0$

$$\Rightarrow V_A - 16 + V_A + 2V_A - 2V_B = 0$$

$$\Rightarrow 4V_A - 2V_B = 16$$

$$\Rightarrow 2V_A - V_B = 8 \quad \text{--- (1)}$$

Applying KCL at node B :-

$$\frac{V_B - V_A}{2} + \frac{V_B}{8} - 4 = 0$$

$$4V_B - 4V_A + V_B - 32 = 0$$

$$5V_B - 4V_A = 32 \quad \text{--- (2)}$$

$$-2V_B + 22V_A = 8 \times 2 \quad \text{--- (1) } \times 2$$

$$\begin{array}{rcl} 3V_B & = & 48 \quad \Rightarrow V_B = \frac{48}{3} V \\ & & = 16 V \end{array}$$

$$V_A = (8 + 16)/2 = 12 V$$

direction of current in 2Ω resistor: $B \rightarrow A$

$$\text{current value} = \frac{V_B - V_A}{2} = 2 \text{ A}$$

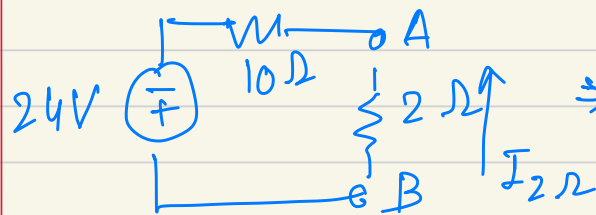
$$A) \text{ Power dissipated} = I^2 R = 4 \times 2 = 8 \text{ W}$$

Alternate solution for part A:- We can use 2Ω as a load resistor

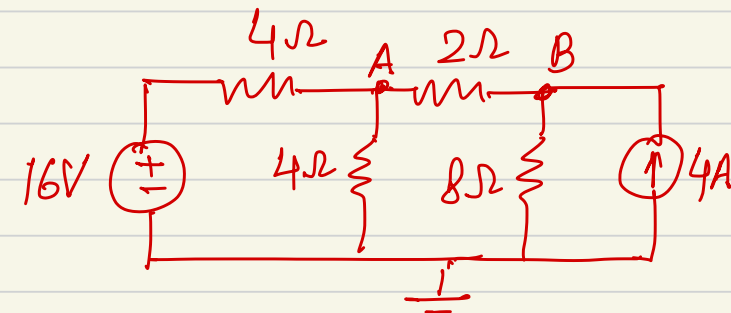
& take the Thevenin approach to calculate $I_{2\Omega}$. Open circuiting 2Ω b/w A & B:-

$$V_A = \frac{4}{8} \times 16 = 8 \text{ V}, \quad V_B = 8 \times 4 = 32 \text{ V}, \quad V_{OC} = V_{AB} = -24 \text{ V}$$

$$R_{AB} \text{ after shorting } 16 \text{ V} \& \text{ opening } 4 \text{ A} \Rightarrow R_{AB} = 4 \parallel 4 + 8 = 10 \Omega$$



$$\Rightarrow I_{2\Omega} = 2 \text{ A} \Rightarrow \text{power dissipated} = I^2 R = 8 \text{ W}$$



Alternatively:- Mesh Analysis

Mesh 1:- $16 - 4I_1 - 4(I_1 - I_2) = 0$

$$4 - I_1 - I_1 + I_2 = 0$$

$$-2I_1 + I_2 + 4 = 0 \quad \text{--- (1)}$$

Mesh 2:-

$$4(I_1 - I_2) - 2I_2 - 8(I_2 - I_3) = 0$$

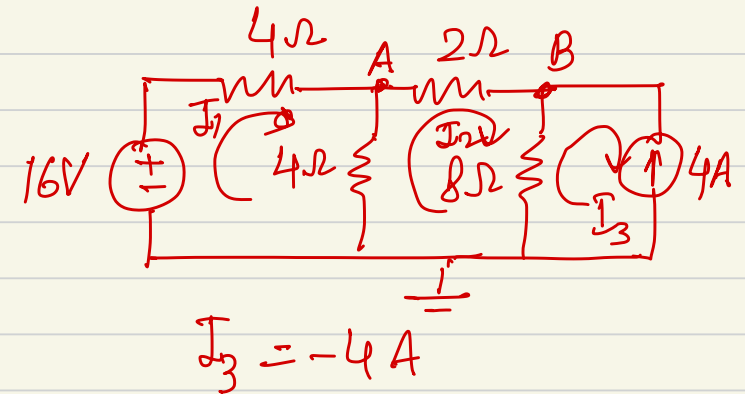
$$2I_1 - 2I_2 - I_2 - 4I_2 + 4I_3 = 0$$

$$2I_1 - 7I_2 - 16 = 0 \quad \text{--- (2)}$$

$$-2I_1 + I_2 + 4 = 0 \quad \text{--- (1)}$$

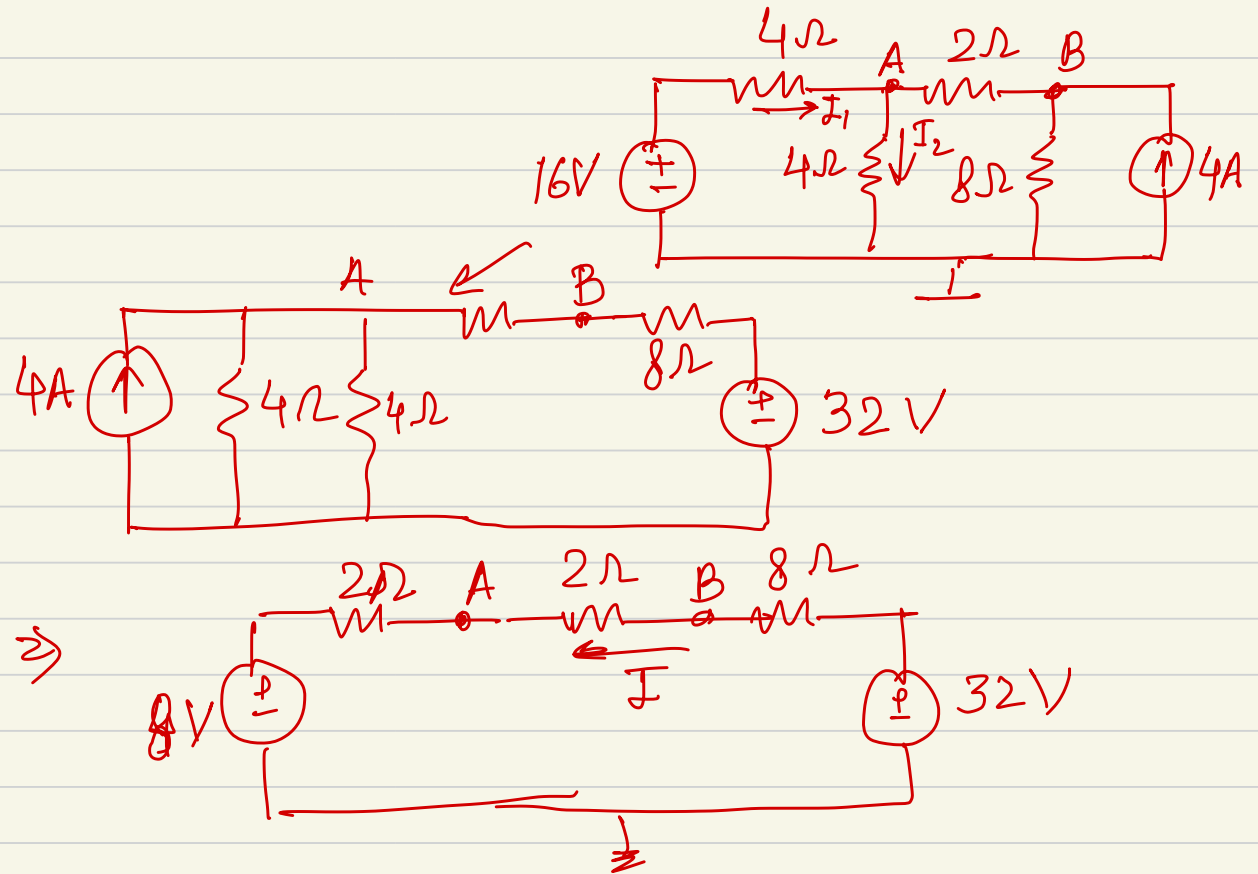
$$-6I_2 - 12 = 0 \Rightarrow I_2 = -2A, \text{ i.e. } I_{B \rightarrow A} = 2A$$

$$\textcircled{1} \Rightarrow 2I_1 = I_2 + 4 = 2 \Rightarrow I_1 = 1A, \text{ from } 16V \text{ source}$$



Alternatively:-

Source transformation



$$\bar{I} = \frac{32-8}{12} = \frac{24}{12} \text{ A} = 2 \text{ A}$$

from B $\rightarrow$ A

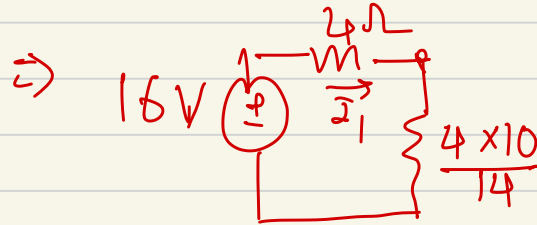
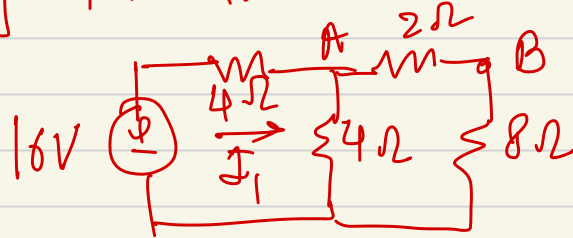
$$V_A = 8 + 2 \times 2 = 12 \text{ V}$$

$$\Rightarrow I_1 = \frac{4}{4} = 1 \text{ A}$$

Alternatively:-

Superposition:-

deactivating 4A source:-

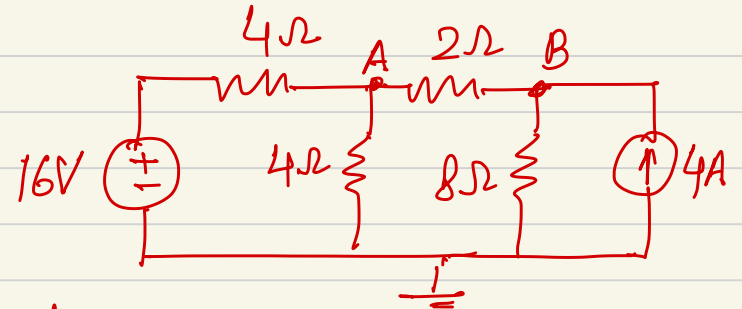


$$I_1 = \frac{16}{4 + \frac{4 \times 10}{14}}$$

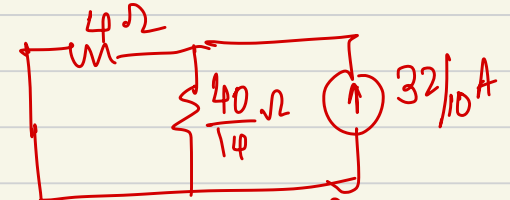
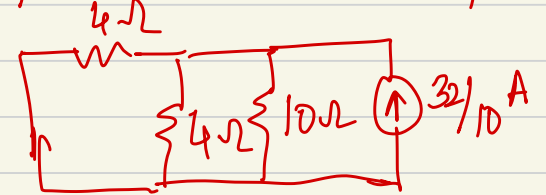
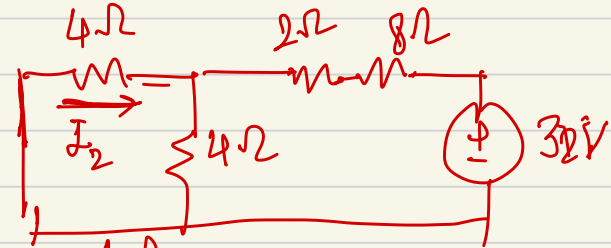
Current flowing through 4Ω resistor in series with

$$16V \Rightarrow I_1 + I_2 = \frac{16 - \frac{32 \times 42}{147}}{4 + \frac{4020}{147}} A$$

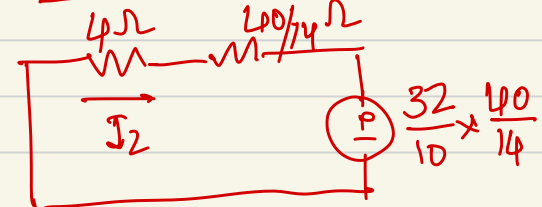
$$= \frac{16 \times 7 - 32 \times 2}{48} = \frac{16 \times 3}{48} = 1 A$$



deactivating 16V source

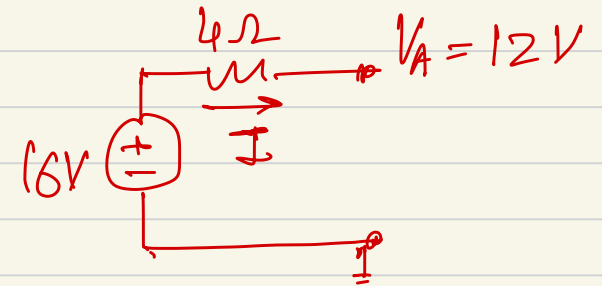


$$\Rightarrow I_2 = -\frac{\frac{32}{10} \times \frac{4}{14}}{4 + \frac{40}{14}}$$



B. To comment on the current flowing in $16V$ branch, we use results from nodal analysis:-

$$I = \frac{16 - 12}{4} = 1A$$

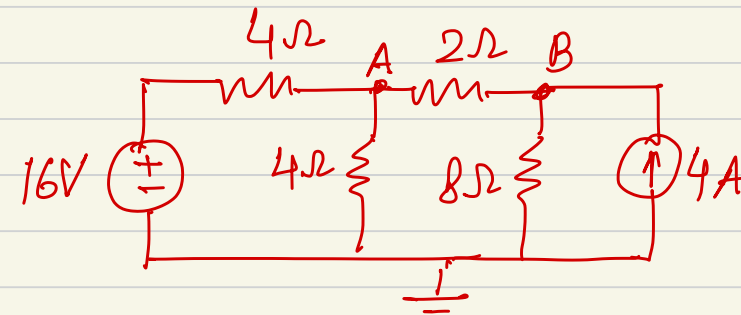


→ delivering the power as it is not flowing inside the source

C. For $V_A = V_B$ to have zero current in 2Ω resistor, we can look at the following approach.

→ We can use 2Ω as a load resistor

& take the Thevenin approach to calculate $I_{2\Omega}$. Open circuiting 2Ω b/w A & B:-

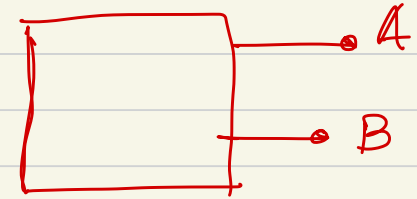
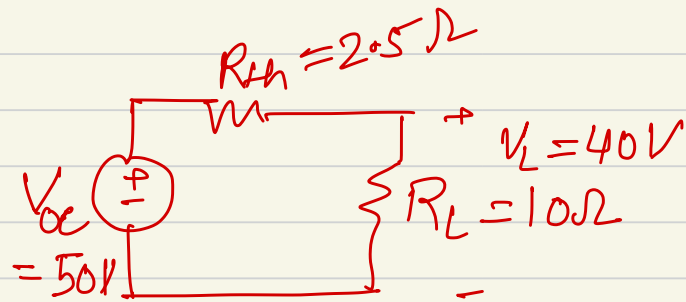


$$V_A = \frac{4}{8} \times 16 = 8V, \quad V_B = 8 \times 4 = 32V,$$

⇒ $I_{\text{load}} = 1A$ if $4A$ becomes $1A \Rightarrow V_B = 8V \Rightarrow V_{AB} = 0$

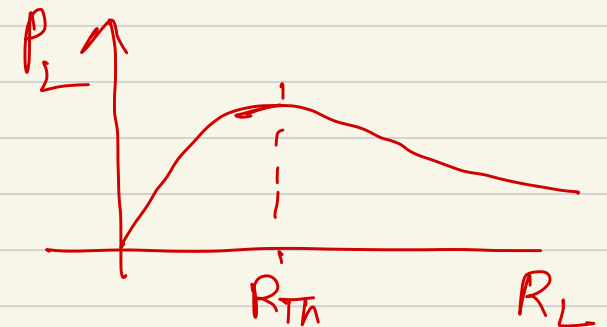
Q3. i) $V_{AB} = 50V = V_{oc}$

ii)



A) $I_L = \frac{40}{10} = 4A \Rightarrow \frac{V_{oc} - V_L}{R_{th}} = 4A \Rightarrow \frac{10V}{R_{th}} = 4A$
 $\Rightarrow R_{th} = \frac{5}{2} \Omega$

B) For maximum power, $R_L = R_{th}$
 (MPT) $= 2.5 \Omega$



C) Total power dissipation at MPT, $= I_L^2 (R_L + R_{th})$, $I_L = 10A$
 in the circuit (internal)
 $= 10^2 \times 5 = 500W$

→ very high power dissipation compared to 50W requirement.

Alternative C part:-

(For people who have take total internal power dissipated in only R_{th} rather than both R_L & R_{th})

$$\text{Total internal PD} = I_L^2 R_{th} = 10^2 \times 2.5 = 250 \text{ W}$$

→ higher than the 50 W requirement.

Alternative D)

$$PD = \frac{V_{oc}^2}{(R_L + R_{th})^2} \cdot R_{th} = 50 \text{ W}$$

$$\frac{50 \times 50}{(R_L + 2.5)^2} = \frac{500}{2.5} \Rightarrow \frac{25 \times 50}{2} = (R_L + 2.5)^2 \Rightarrow R_L = 5\sqrt{5} - 2.5$$

$$E) \text{ Efficiency} = \frac{\left(\frac{50}{R_L + R_{th}} \right)^2 R_L}{50} =$$



$$D) \quad P_{D_{tot}} = \frac{V_{oc}^2}{(R_L + R_{th})^2} \times (R_L + R_{th}) = \frac{V_{oc}^2}{R_L + R_{th}}$$

For meeting the target requirement of 50 mW = $\frac{(50)^2}{R_L + 2.5}$

$$\Rightarrow R_L + 2.5 = 50 \Rightarrow \underline{R_L = 47.5 \Omega}$$

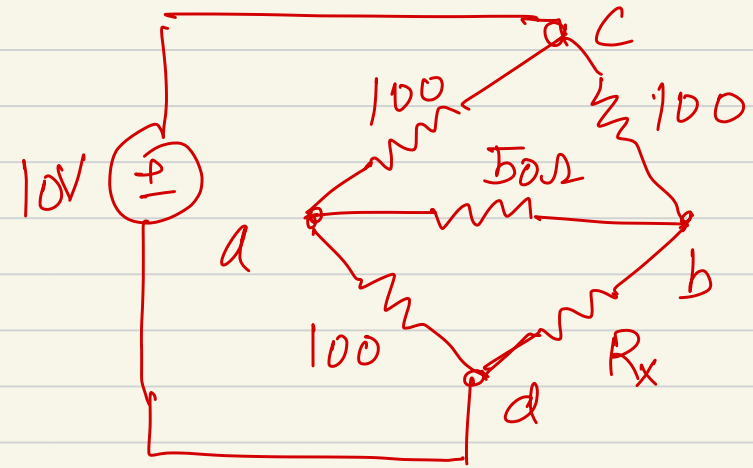
E) What is efficiency at this point

$$= \frac{\left(\frac{50}{50}\right)^2 \times 47.5}{50} = \underline{\frac{47.5}{50}}$$

Q.4

When $R_x = 100\Omega$, the bridge is in balanced condition i.e.

$V_A = V_B \Rightarrow$ current flowing through 50Ω resistor is zero.



During impact, $R_x = 80\Omega$,

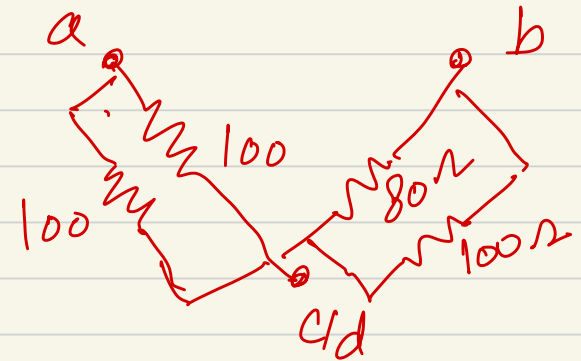
bridge is unbalanced now and we want to calculate the voltage & current 50Ω resistor.

As 50Ω can be treated as load resistor connected between a & b :- we can open circuit a & b and calculate the Thevenin equivalent b/w a & b:

$$V_a = 10 \times \frac{100}{200} = 5V, \quad V_b = 10 \times \frac{80}{180} = \frac{40}{9}V$$

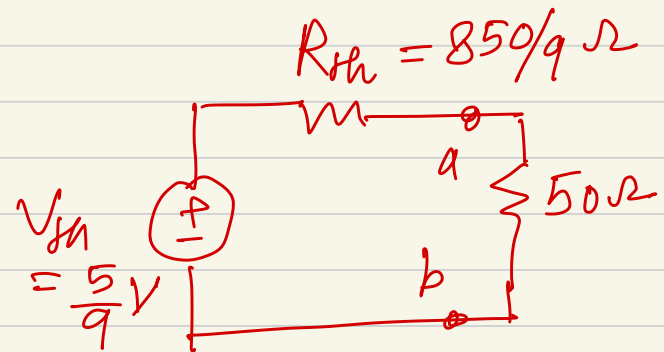
∴ R_{th} , we can short c & d , i.e. $10V$ deactivated.

$$\begin{aligned}\Rightarrow R_{ab} = R_{th} &= 100 \parallel 100 + 100 \parallel 80 \\ &= 50 + \frac{100 \times 80}{180} \\ &= 50 + \frac{400}{9} = \frac{850}{9} \Omega\end{aligned}$$



$$V_{th} = 5 - \frac{40}{9} = \frac{5}{9} V$$

$$I_{ab} = \frac{5/9}{50 + \frac{850}{9}} = \frac{5}{1300} A$$



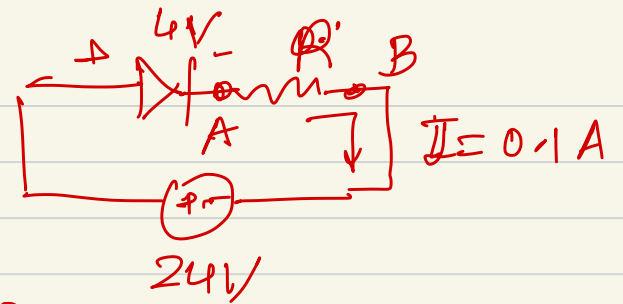
$$V_{out} = I_{ab} \times 50 = \frac{250}{1300} = \frac{255}{13026} = \frac{5}{26} V$$

Q5. Given

required $\Rightarrow R = \frac{24-4}{0.1} = 200\Omega$

Power dissipation in $R = I^2 R = (0.1)^2 \cdot 200$
 $= 2W$

$\rightarrow$ more than what it can handle.



i.e. if I want to limit current flowing through the resistance/resistive network as $0.1A$ directly flowing through a 200Ω resistor is causing it to burn. I only have 200Ω resistor available.

$\Rightarrow R_{eq} = 200\Omega$, & Power Diss in each resistor $\leq 0.5W$

$1/4$ lower than what we are getting.

$\Rightarrow$ Decreasing current by $1/2$ can get

us this rating $\rightarrow$ parallel network
but we don't have 400Ω , only 200Ω

$\Rightarrow 200\Omega$ in series and two branches in parallel

